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Fuzzy Logic Based Threat Assessment Application In Air Defense Systems

Muhittin Coskun and Sakir Tasdemir

Abstract— **In this study, a fuzzy logic based threat assessment application is presented. In order to create the application in the most accurate way, the help of air defense experts is taken. First of all, speed, distance, altitude, radar cross section, jamming signal level and distance of the target to the closest point of approach parameters are taken as input parameters and the threat degree is taken as output parameter in order to obtain the threat degrees of air targets. Then, an inference mechanism containing 133 rules is created, allowing the right threat degrees to be assigned to any kind of air target. Moreover, unlike other fuzzy logic-based threat assessment applications, air targets, which are the biggest threats to air defense systems such as ballistic missiles and cruise missiles, have been included in test scenarios and inference system foundations to ensure the system's excellence. In addition, by using the radar cross section parameter, which is not used in similar studies, it is ensured that the application can distinguish between air targets with approximate speed, altitude and distance values, and the deficiency in the literature is eliminated.**

***Index Terms*—Fuzzy logic systems, threat assessment, air defense.**

I. Introduction

IN the modern air warfare environment, it is a vital issue for air defense systems to evaluate (analyze) a wide variety of types and numbers of air targets and to make a threat rating and to eliminate the threat by ensuring that these threats are allocated to appropriate weapon systems.

The Air Defense Threat Evaluation (TE) process is an extremely complex process and can only be performed by experienced and skilled experts in the field. Since air targets are dynamic and uncooperative threats, it is vital for both air defense systems and the defended area to evaluate them quickly and put in place appropriate counter-attacks [1]. Air defense personnel are often engaged in difficult assignments. In addition to being responsible for all aircraft in the surveillance area, they should raise awareness of available resources, monitor voice and verbal messages, and prepare status reports. In this environment, it may be difficult for air defense team members to recognize or identify important information that may enable them to better understand the tactical situation [2]. Therefore, a model which will support those who try to solve air warfare puzzle with a prioritization of air targets is actually crucial for survival of an air defense system.

At times, threat assessment becomes difficult in the presence of multiple parameters and processes. There is some uncertainty in these parameters depending on the nature of the objectives and assets involved. It is difficult to formulate the mathematical model using the parameters chosen as input to generate the threat value as an output. At this point, fuzzy logic system emerges as one of the most effective methods for threat assessment of dynamic targets under uncertain conditions [3].

Although there are many methods for air target threat assessment in the literature, such as multi-attribute decision-making method, multi-criteria compromise solution sorting method, fuzzy logic method, rough set theory, Bayesian network and so on, in this article, air threat assessment is performed by using fuzzy logic method[4]-[9]. Because the main problem involved in the air defense information fusion process is the need to deal with uncertainty [10]. And the one of the best ways to deal with uncertainty is fuzziness [3].

Various parameters are needed to show the correlation between targets and protected assets for threat assessment for air defense systems. Different parameters have been proposed in various studies and these are generally divided into proximity, capacity and intention parameters [11]. Proximity parameters, in a sense, are the parameters that measure the proximity of the target to the defended asset and/or the air defense system. A key parameter used in many threat assessment techniques is the distance from the defended asset to the closest point of approach (CPA) [12]. Capacity parameters reflect the ability of an air target to inflict damage to a defended asset. The central parameter here is the target type. Since the target type is often not known with certainty, the target's estimated velocity, radar cross section (RCS), response to the IFF interrogation can give a reasonable indication of what type of target has been detected. The parameters given above are critical, as the target type cannot be clearly known [4]. The intent parameters class is a broad category that includes parameters that can reveal something about the target's intent to harm the defended asset. An example of this is the kinematics of the target. The speed (i.e., speed and direction) of the target,

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along with its altitude, can be a good indicator of the target's intent to attack a defended entity. The uses of radar jamming and electronic warfare techniques are also included in this parameter class [1].

As stated above, although many parameters are used for threat assessment in air defense systems [13], we have included 6 parameters in our study with which we can reach the right conclusion as quickly as possible. The input parameters we used in this article are *speed (km/h), distance (km), altitude (km), radar cross-section (RCS) (m²), jamming signal level (JSL) (dBm), closest point of approach (CPA) (m),* and the output parameter is *Threat Degree (%).*

When other fuzzy logic-based air threat assessment studies in the literature are examined [14]-[19], it is seen that these studies do not include ballistic missiles and cruise missiles, which are major threats to air defense systems [20]-[21]. In addition, it was observed that the RCS parameter, which allows to distinguish between air targets which have similar speed, altitude and range parameters is missing [22]. However, ballistic missiles which have very small RCS values and very high speeds are very difficult to detect by air defense systems and are among the most dangerous air targets due to their short reaction time [23]-[25]. In addition, cruise missiles with very low RCS values and advancing to the target point by making use of the ground are difficult to detect by air defense systems, and their reaction times are short as they can only be detected from close distances [25]. Therefore, ballistic missiles, cruise missiles and the RCS parameter must be taken into account when designing a threat assessment application for air defense systems for use in a modern air warfare environment. In this study, a fuzzy logic-based threat assessment application is proposed to ensure the survivability of air defense systems by eliminating the aforementioned deficiencies. With the use of the proposed application by air defense systems, human errors that can be made in the short reaction time caused by the nature of air defense activities are prevented and the survival of air defense systems is ensured by making the most accurate threat assessment even in complex air warfare environments.

II. Fuzzy Sets Theory

This theory was first put forward by Lotfi Ali Asker-Zadeh, an American of Azeri origin, in 1965 [26]. In his article known as fuzzy sets, Zadeh showed that he could relate language and human intelligence and proposed the theory of fuzzy sets for this. Zadeh has shown that many concepts can be defined linguistically better than traditional mathematics and that fuzzy logic and its sets better model real life.

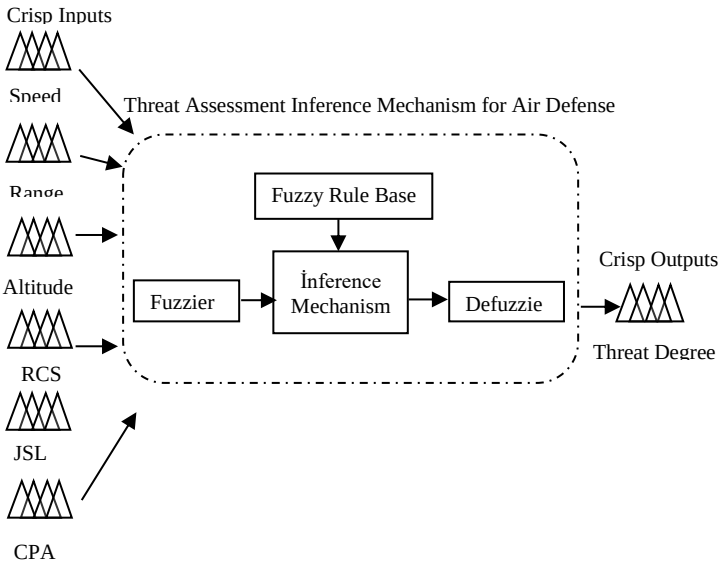
A fuzzy set F defined on a universe of discourse U is characterized by a membership function μF(x) that takes values in the range [0, 1]. A fuzzy set is a generalization of an ordinary subset (i.e., a crisp subset) whose membership function takes only two values, either zero or unity. A membership function provides a measure of the degree of similarity of an element in U to the fuzzy subset [27]. A triangular membership function μF with endpoints (a, 0) and (0, b), and high point (c, α) which we used in our application is characterized [28] as in (1):

$$\mu F(x) = \begin{cases} \alpha\left(\frac{x-a}{c-a}\right) & a \le x \le c \\ \alpha\left(\frac{x-b}{c-b}\right) & c \le x \le b \\ 0, & otherwise \end{cases} \quad (1)$$

Fuzzy rule-based expert systems are one of the most common uses of fuzzy logic. Decision making is an important part of the entire system [29]. A basic fuzzy inference system may take either uncertain inputs or clear inputs, but the outputs it produces are almost always fuzzy sets. When the inference system is used as a controller, it is necessary to have a clear output. Therefore, an intelligent system is created by utilizing the knowledge and experience of experts and/or experts [30]. In this system, there are membership functions created using linguistic variables, if-then rules and inference mechanism [31].

III. Fuzzy Logic Based Threat Assessment Application in Air Defense Systems

Our model was designed by the help of air defense experts. The proposed application, which is prepared by the help of air defense experts, is designed to make an accurate, rapid and human error-free threat assessment against all types of air targets, thus increasing the success of air defense systems even in complex air warfare environments. This fuzzy rule-based inference system is defined as the Fuzzy Expert System (FES) [32]. For the proposed FES, the speed, distance, RCS, JSL, and CPA parameters of the air targets were determined as the input parameters and the threat degree as the output value. The values of input parameters were created by the help of experts to simulate the real air targets and translated into fuzzy sets with linguistic expressions. Then fuzzy rules were created by the help of experts and run in the rules inference mechanism using the Mamdani method. As a result, the fuzzy outputs were defuzzied to obtain precise threat levels. For the process, MATLAB is used. The general structure of the proposed FES is shown in Fig. 1.



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Fig. 1. General structure of proposed fuzzy logic-based threat assessment application

In order to test the proposed fuzzy expert system, air threats based on air targets characteristics are created by the help of the experts and presented in TABLE I. The parameters of the mentioned targets have been created in order to simulate the real situations by considering the target characteristics [33]-[39]. The test scenarios presented in TABLE I are created to include any air target that an air defense system could encounter. By creating 2 scenarios with different parameters from each type of air target, it is aimed to show that the proposed application can accurately evaluate air targets with various parameters.

TABLE I
TEST SCENARIOS

No	Target Type	Speed (km/h)	Range (km)	Altitude (km)	RCS (m^2)	JSL (dBm)	CPA (m)
1	Helicopter	300	75	2	0,9	0	40
2	Commercial Airplane	800	90	10	9	0	250
3	Jamming Aircraft	700	120	7	4	7	0
4	Fighter	800	60	2,5	3	0	5
5	Bomber	750	130	11	13	0	0
6	Unmanned Air Vehicle Aircraft	600	80	4,5	0,6	0	25
7	Launched Missile	1100	65	5	0,08	0	10
8	Cruise	650	40	0,5	0,1	0	15
9	Ballsitic Missile	3000	55	28	0,15	0	20
10	Fighter 2	700	120	4	4	0	10
11	Helicopter2	300	25	2	0,9	0	40
12	Commercial Airplane 2	800	60	10	9	0	120
13	Jamming Aircraft 2	700	120	7	4	4	0
14	Bomber 2	600	130	11	13	0	90
15	Unmanned Air Vehicle 2 Aircraft	600	150	4,5	1,5	0	25
16	LAunched Missile 2	1100	90	5	0,08	0	80
17	Cruise Missile 2	650	75	1	1	0	15
18	Ballistic Missile 2	1500	55	6	0,15	0	20

The equations limiting the input parameters and out parameter used in the FES are as in (2)-(8):

$$Speed(x) = \begin{cases} x, & 0 \le x \le 3000 \\ 0, & Otherwise \end{cases} \quad (2)$$

$$Distance(y) = \begin{cases} y, & 0 \le y \le 30 \\ 0, & Otherwise \end{cases} \quad (3)$$

$$Altitude(z) = \begin{cases} z, & 0 \le z \le 30 \\ 0, & Otherwise \end{cases} \quad (4)$$

$$RCS(t) = \begin{cases} t, & 0 \le t \le 20 \\ 0, & Otherwise \end{cases} \quad (5)$$

$$JSL(u) = \begin{cases} u, & 0 \le u \le 10 \\ 0, & Otherwise \end{cases} \quad (6)$$

$$CPA(v) = \begin{cases} v, & 0 \le v \le 500 \\ 0, & Otherwise \end{cases} \quad (7)$$

$$TD(k) = \begin{cases} k, & 0 \le k \le 100 \\ 0, & Otherwise \end{cases} \quad (8)$$

A. Memberships Functions

We labeled each input and output parameter by a set of three to six fuzzy characteristics sets as shown in TABLE II.

TABLE II
LINGUISTIC VARIABLES OF PARAMETERS

Parameter	MFs
Speed	Very Slow (VS), Slow (S), Fast (F), Very Fast (VF), Very Very Fast (VVF)
Range	Close (C), Far (F), Very Far (VF)
Altitude	Low (L), Medium (M), High (H), Very High (VH)
RCS	Very Small (VS), Medium (M), Big (B), Very Big (VB)
JSL	Little (L), Much (M), Very Much (VM)
CPA	Very Close (VC), Close (C), Far (F),
TD	Very Low (VL), Low (L), Medium (M), High (H), Very High (VH), Very Very High (VVH)

We used triangular membership functions as stated in (1) and each membership function is decided by the help of aforementioned experts to be able to achieve the best accurate results. Graphics belonging to membership functions of the input parameter Speed are shown in Fig. 2. Other membership functions of the other parameters are similar to the one shown in Fig. 2.

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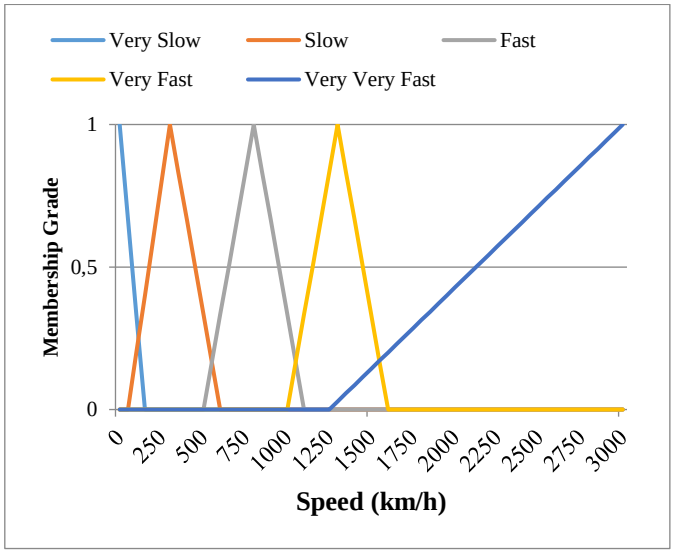


Fig.2. MFs of the speed

Membership functions of the input parameter Speed (x) are described in (9)-(13) where x is the member of fuzzy set Speed (x) and μSpeed(x) is the degree of membership of x in Speed (x):

$$\mu VS(x) = (x;\ 0{,}150) = \begin{cases} \frac{150-x}{150}, & 0 \le x < 150 \\ 0, & x \ge 150 \end{cases} \quad (9)$$

$$\mu S(x) = (x;\ 50{,}300{,}600) = \begin{cases} 0, & x \le 50 \\ \frac{x-50}{300-50}, & 50 < x \le 300 \\ \frac{600-x}{600-300}, & 300 < x < 600 \\ 0, & x \ge 600 \end{cases} \quad (10)$$

$$\mu F(x) = (x;\ 500{,}800{,}1100) = \begin{cases} 0, & x \le 500 \\ \frac{x-500}{800-500}, & 500 < x \le 800 \\ \frac{1100-x}{1100-800}, & 800 < x < 1100 \\ 0, & x \ge 1100 \end{cases} \quad (11)$$

$$\mu VF(x) = (x;\ 1000{,}1300{,}1600) = \begin{cases} 0, x \le 1000 \\ \frac{x-1000}{1300-1000}, 1000 < x \le 1300 \\ \frac{1600-x}{1600-1300}, 1300 < x < 1600 \\ 0, x \ge 1600 \end{cases} \quad (12)$$

$$\mu VVF(x) = (x;\ 1250{,}3000) = \begin{cases} 0, x \le 1250 \\ \frac{x-1250}{3000-1250}, 1250 < x \le 3000 \\ 0, x > 3000 \end{cases} \quad (13)$$

Other membership functions of the input and output parameters are similar to the one shown in equations (9)-(13).

B. *Fuzzy Rules*

The fuzzy inference rules created by the help of the aforementioned experts to determine the most accurate threat degree for air targets are shown in TABLE III. By using 133 fuzzy inference rules in our application, it is aimed for air defense systems to successfully evaluate all kinds of air threats.

In addition, as stated above, while the rules are created in order to design a powerful algorithm by the help of air defense experts, the most modern and powerful air threats are taken into account, ensuring that the study is novel and the application is flawless. To be able to ensure the novelty of the study, the RCS parameter, which is not used in similar studies, but is very important for target discrimination in air defense systems, is used effectively while creating the rules. Furthermore, ballistic missiles, cruise missiles and the targets launched from the aircrafts, which are the most dangerous air targets for air defense systems [20]-[25], are also taken into account in the rules, unlike similar studies, so the algorithm is guaranteed to be flawless.

TABLE III
FUZZY RULES

No	Rule
1	If (Speed(km/h) is VVF) and (Range(km) is C) and (Altitude(km) is VH) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is VVH)
2	If (Speed(km/h) is VF) and (Range(km) is C) and (Altitude(km) is M) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is VVH)
3	If (Speed(km/h) is VF) and (Range(km) is F) and (Altitude(km) is M) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is VH)
4	If (Speed(km/h) is VF) and (Range(km) is VF) and (Altitude(km) is M) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is H)
5	If (Speed(km/h) is VF) and (Range(km) is C) and (Altitude(km) is H) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is VH)
6	If (Speed(km/h) is VF) and (Range(km) is C) and (Altitude(km) is VH) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is H)
7	If (Speed(km/h) is VF) and (Range(km) is C) and (Altitude(km) is M) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is C) then (TD is VH)
8	If (Speed(km/h) is VF) and (Range(km) is C) and (Altitude(km) is M) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is F) then (TD is H)
9	If (Speed(km/h) is VF) and (Range(km) is C) and (Altitude(km) is M) and (RCS(m) is VS) and (JSL(dBm) is L) and (CPA(m) is VF) then (TD is M)
…..	……
…..	……
131	If (Speed(km/h) is F) and (Range(km) is F) and (Altitude(km) is L) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is VH)
132	If (Speed(km/h) is F) and (Range(km) is C) and (Altitude(km) is M) and (JSL(dBm) is L) and (CPA(m) is VC) then (TD is VH)
133	If (Speed(km/h) is VVF) and (RCS(m) is VS) and (CPA(m) is VC) then (TD is VVH)

C. *Inference Mechanism and Defuzzification*

The inference mechanism checks each created rule and chooses which rule to run based on the fuzzy inputs. When the system is started by choosing the Mamdani max-min method due to its simplicity and ease of calculation as a fuzzy inference mechanism [40], the accuracy degrees (α) will be shaped as follows:

Rule 1- α_1= min(μVVF(x), μC(y), μVH(z), μVS(t), μL(u), μVC(v))

Rule 2- α_2= min(μVF(x), μC(y), μM(z), μVS(t), μL(u), μVC(v))

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Rule 3- α_3= min(μVF(x), μF(y), μM(z), μVS(t), μL(u), μVC(v))

Rule 4- α_4= min(μVF(x), μVF(y), μM(z), μVS(t), μL(u), μVC(v))

Rule 5- α_5= min(μVF(x), μC(y), μH(z), μVS(t), μL(u), μVC(v))

Rule 6- α_6= min(μVF(x), μC(y), μVH(z), μVS(t), μL(u), μVC(v))

Rule 7- α_7= min(μVF(x), μC(y), μM(z), μVS(t), μL(u), μC(v))

Rule 8- α_8= min(μVF(x), μC(y), μM(z), μVS(t), μL(u), μF(v))

Rule 9- α_9= min(μVF(x), μC(y), μM(z), μVS(t), μL(u), μVF(v))

…….

…….

…….

Rule 131- α_{131}= min(μF(x), μF(y), μL(z), μL(u), μVC(v))

Rule 132- α_{132}= min(μF(x), μC(y), μM(z), μL(u), μVC(v))

Rule 133- α_{133}= min(μVVF(x), μVS(t), μVC(v))

When the rules are run, their minimum values are taken first. However, if more than one rule is run, the maximum accuracy of each rule is taken and the result is calculated as in (14).

$$\alpha_{1,2,\ldots,n} = \max(\alpha_1, \alpha_2, \ldots, \alpha_n) \quad (14)$$

For instance; when a target parameters with Speed=700 km/h, Distance=20 km, Altitude=4 km, RCS=4 m², JS=7 dBm and CPA=0 m are entered, Rule 67 and 73 will work. In this case, the degree of accuracy is calculated as follows:

α_{67}= min(μF(x), μF(y), μM(z), μM(t), μVM(u), μVC(v))

α_{67}= min(0.67, 1, 0.33, 0.5, 1, 1)

α_{67}= min(0.33)

α_{73}= min(μF(x), μF(y), μH(z), μM(t), μVM(u), μVC(v))

α_{73}= min(0.67, 1, 0.4, 0.5, 1, 1)

α_{73}= min(0.4)

$\alpha_{67,73}$=max (0.33, 0.4)

$\alpha_{67,73}$= 0.4

A fuzzy output is obtained from the fuzzy inference mechanism. The process of converting this output to a crisp output is defuzzification. For each rule used, a fuzzy output set consisting of membership values is obtained. By using one of the methods of defuzzification on the logical combination set of the fuzzy output set, the defuzzification process is performed. In the above example, the accuracy level is 0.4. In the defuzzification process, the centroid method [41] given in (15) was used with the degree of accuracy found;

$$x' = \frac{\sum_{i=a}^{b} \mu A(xi)xi}{\sum_{i=a}^{b} \mu A(xi)} \quad (15)$$

where μA(xi) is a membership grade in MF, xi is a sample element and b represents the number of samples in the fuzzy set A . The exact result from this expression was found TD=69.1 in MATLAB.

IV. RESULTS

The fuzzy logic based threat assessment application presented in this article has been tested in the scenarios given in Table I. In order to determine the threat degrees of air targets, 6 input parameters belonging to targets are fuzzied as stated above, and the threat assessment process is started. Then, fuzzy results are obtained by the inference mechanism created within the framework of the rules prepared by the help of experts, and the Threat Degrees are obtained by clarification.

The test results prepared with 18 air targets are presented in Table IV. In the test simulation, 2 scenarios were created for each target type and it was shown that the proposed application successfully assess all kinds of targets and various parameters. The Threat Degree is directly proportional to the danger of the air target.

It has been observed that the proposed application assigns a higher threat degree to the toughest targets due to their destructive power, short reaction times and difficulties in detection for air defense systems such as ballistic missiles, cruise missiles and aircraft-launched missiles. When the other threat degrees are examined together with the air defense experts, it has been seen that the target parameters and the target threat levels are compatible in terms of ensuring the survivability of the air defense systems.

TABLE IV
THREAT DEGREES OBTAINED BY FUZZY LOGIC BASED THREAT ASSESSMENT APPLICATION

No	Target Type	Speed km/h	Range km	Altitude km	RCS m²	JSL dBm	CPA m	TD %
1	Helicopter	300	75	2	0,9	0	40	**42,2**
2	Commercial Airplane	800	90	10	9	0	250	**7,52**
3	Jamming Aircraft	700	120	7	4	7	0	**69,1**
4	Fighter	800	60	2,5	3	0	5	**72,3**
5	Bomber	750	130	11	13	0	0	**54,5**
6	Unmanned Air Vehicle Aircraft	600	80	4,5	0,6	0	25	**64,4**
7	Launched Missile	1100	65	5	0,08	0	10	**84,6**
8	Cruise	650	40	0,5	0,1	0	15	**92,5**
9	Ballsitic Missile	3000	55	28	0,15	0	20	**93,4**
10	Fighter 2	700	120	4	4	0	10	**61,6**
11	Helicopter2	300	25	2	0,9	0	40	**60**
12	Commercial Airplane 2	800	60	10	9	0	120	**15,9**
13	Jamming Aircraft 2	700	120	7	4	4	0	**56,5**

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No	Target Type	Speed km/h	Range km	Altitude km	RCS m²	JSL dBm	CPA m	TD %
14	Bomber 2	600	130	11	13	0	90	**48,9**
15	Unmanned Air Vehicle 2	600	150	4,5	1,5	0	25	**51,2**
16	Aircraft Launched Missile 2	1100	90	5	0,08	0	80	**67,4**
17	Cruise Missile 2	650	75	1	1	0	15	**80,2**
18	Ballistic Missile 2	1500	55	6	0,15	0	20	**83,4**

The multitude and diversity of air targets in the modern air warfare environment make the work of air defense systems very difficult. The threat degree obtained through the application presented in this article will ensure the survivability of air defense systems by enabling those who command air defense operations to notice high-value targets during intense air operations.

V. Conclusion

In this article, a fuzzy logic-based threat assessment application is presented. By using crucial 6 input and 1 output parameters in the application, it is ensured that the most accurate threat ratings of air targets are obtained. By including the RCS parameter, which is not used in other fuzzy logic-based applications, in the study, it was possible to distinguish between targets which have approximate speed, range, and altitude values, and the deficiency in the literature is eliminated. In addition, the reliability of the application has been proven by adding air targets, which are very important targets for air defense systems such as ballistic missiles and cruise missiles, to the test scenarios.

With the our proposed application, air defense systems operating in especially complex air warfare environments will have an important gain in ensuring the survival of the area they protect with their own survival, by making threat prioritization both quickly and accurately and by avoiding human error.

It is considered that it will be possible to make the application completely perfect if on-site studies are carried out with air defense systems. In addition, the application can be further developed using hybrid artificial intelligence techniques.

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